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Density Functional Theory Surrogate Enables Fast and Broad Computational Evaluation of Homogeneous Transition Metal Catalytic Energy Landscapes

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Integration of Machine Learning with DFT

1. Applicable to almost any transition metal and ligand combination
2. Reasonable balance of speed and accuracy

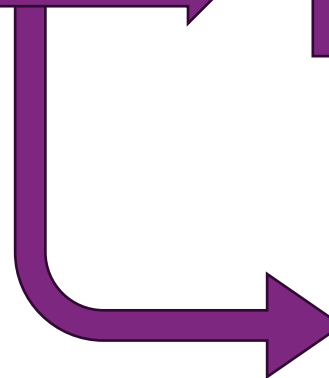
Density Functional Theory
(DFT)

1. Require significant computer resources

1. Approximate density functionals
2. Construct components of density functionals
3. Speed up energy convergence
4. Create better electronic occupation guesses
5. Direct property predictions

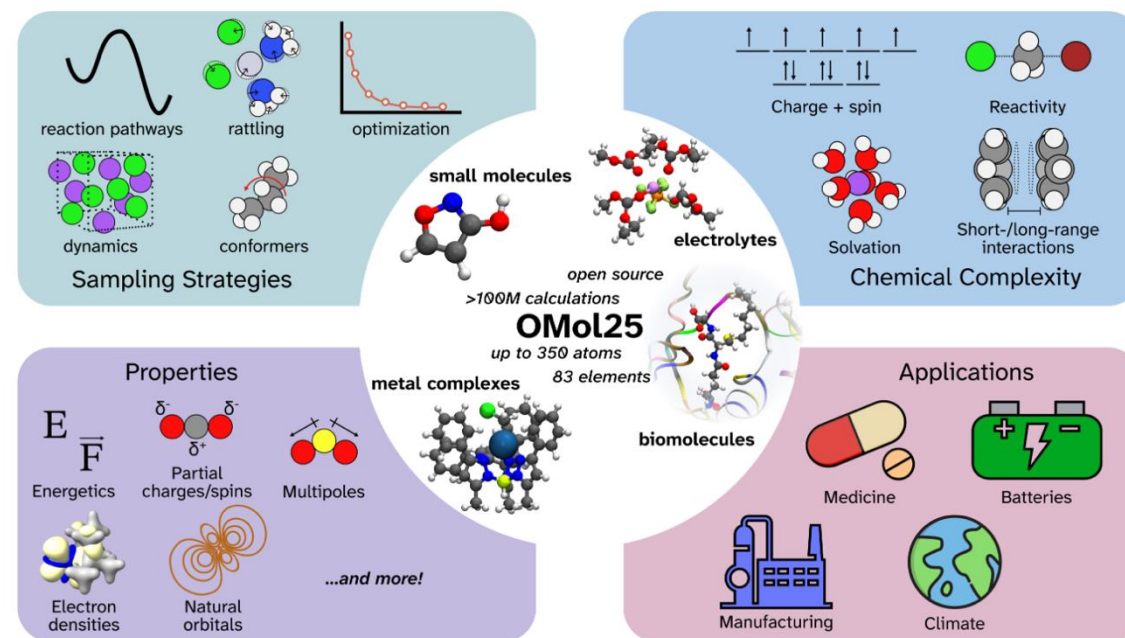
Integration of Machine Learning

Machine Learning Interatomic Potentials
(MLIPs)



MLIPs

- MLIPs are generally trained using reference DFT or *ab initio* structures, energies, and forces.
- MLIPs have the possibility to closely mimic complex potential energy surfaces and provide accurate prediction of energies and forces
- MLIPs have a computational cost much closer to force-field-type calculations than DFT calculations.
- Open Molecules 2025 (OMol25) dataset constructed from >100 million ω B97M-V/def2-TZVPD DFT calculations
- Span areas of chemical space including organic molecules and inorganic/organometallic complexes with a wide variety of charge, spin multiplicities, >80 elements, and with systems of over several hundred atoms.



Visual overview of OMol25

UMA: A Family of Universal Models for Atoms

- A new architecture and set of pre-trained models trained on OMol25 as well as the OC20, ODAC23, OMat24, and forthcoming Open Molecular Crystals 2025 (OMC25) datasets.
- UMA are equivariant Smooth Energy Network (eSEN) based models with different sizes (Small, Medium, and Large).
- UMA feature a mixture of linear experts (MoLE) architecture.
- Retains accuracy and computational efficiency for the various tasks.
- Not explicitly incorporate any charge- or spin-related physics.
- Errors are noticeably higher for systems with net charge or unpaired electrons.

Density Functional Theory Surrogate Enables Fast and Broad Computational Evaluation of Homogeneous Transition Metal Catalytic Energy Landscapes

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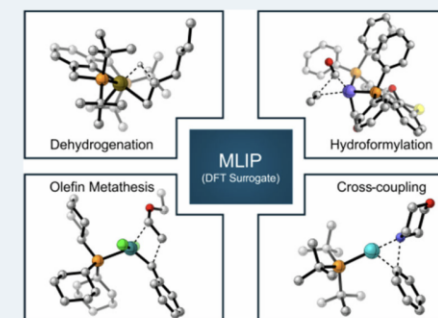
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ABSTRACT: The recently disclosed machine learning interatomic potential (MLIP), universal models for atoms (UMA), due to extensive training, has the potential to act as an extremely fast density functional theory (DFT) surrogate to evaluate the energies and structures of transition metal catalyzed reactions. Here, we report the evaluation and general success of UMA to calculate intermediates and transition structures and energies for catalytic cycles with a variety of metal centers, oxidation states, and ligand frameworks. Examples of catalytic cycles evaluated include alkane dehydrogenation, hydroformylation, olefin metathesis, cross-coupling, and asymmetric reactions. Overall, this evaluation indicates that UMA can be a reasonable surrogate for DFT and UMA energies and structures mimic what can be obtained with the ω B97M-V density functional. However, similar to using the ω B97M-V functional, caution should be taken when applying this MLIP to open-shell and multireference systems. Regardless, it is very likely that UMA and related MLIPs will rapidly be integrated into the computational assessment of homogeneous catalytic cycles.

KEYWORDS: machine learning interatomic potential, density functional theory, dehydrogenation, computational assessment



K. P. Quirion, W.-Y. Kong, B. Stanley, J. Joy, and D. H. Ess, *ACS Catal.*, 2026, **16**, 8053–8066.



- **Aims of the paper:**

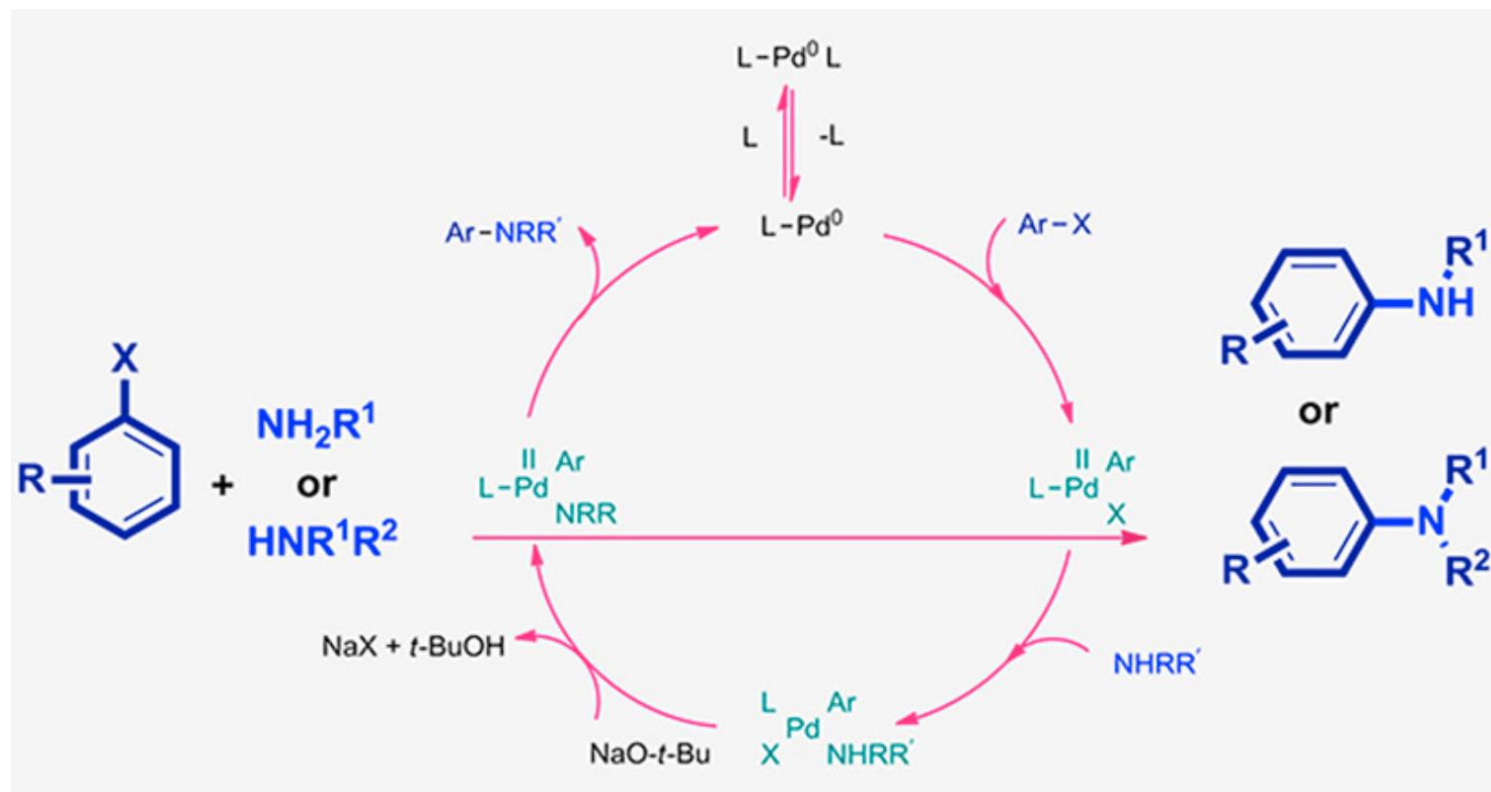
1. Determine if UMA/OMol25 can provide structures and energies commensurate with ω B97M-V50/def2-TZVPD which is a level of theory generally assumed to be broadly applicable and accurate to transition metal reactions.
2. Test the application of UMA for catalytic cycles that involve metal oxidation state changes, changes in coordination number, unique intermediates, and a variety of ligand environments.
3. Describe how homogeneous transition metal catalytic cycles can be rapidly evaluated with a foundation MLIP with reasonable accuracy, which will impact the workflow of future mechanistic studies and catalyst design efforts.

- **Computational Methods:**

1. Single-point energies: Calculated with UMA-S-1.1/OMol25 and UMA-M-1.1/OMol25 on the original DFT-optimized geometries.
2. Geometry optimization: Performed with UMA-S, using the ORCA 6.1.0 optimizer through orca-external-tools
3. Solvation for UMA: For selected reactions, GFN2-xTB/ALPB provided additive corrections to energies and gradients because implicit solvation was absent from UMA's development.
4. DFT reference: UMA was evaluated against its training functional, ω B97M-V, using def2-TZVPD and DEFGRID2 in ORCA.
5. Solvation for DFT: SMD was used when solvent effects were evaluated.

Case study 1: Cross-coupling Catalysis: Pd catalyzed Buchwald-Hartwig amination reaction

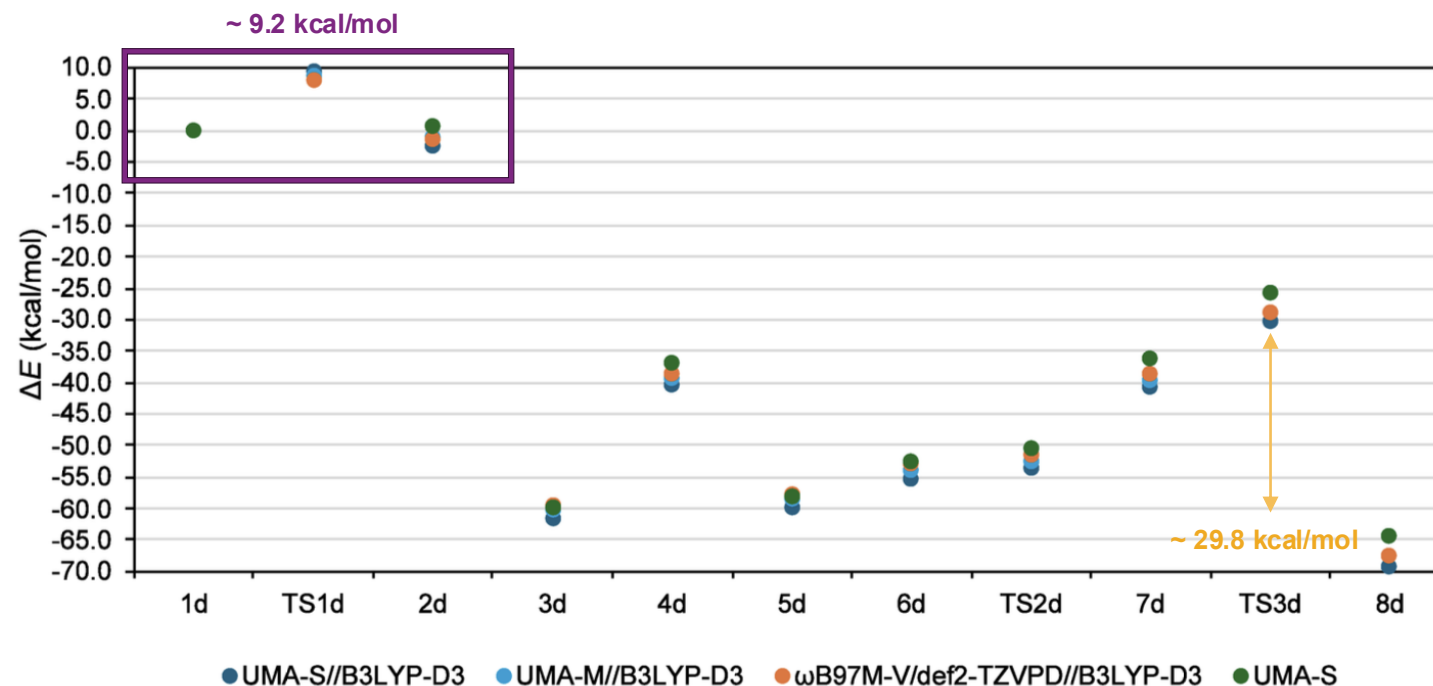
- Forms a carbon-nitrogen (C–N) bond between an amine (1° or 2°) and an aryl halide.
- Mechanism:
 - 1) Oxidative Addition
 - 2) Amine Coordination & Deprotonation
 - 3) Reductive Elimination



Overview of Buchwald-Hartwig amination reaction

Cross-coupling Catalysis: Pd catalyzed Buchwald-Hartwig amination reaction

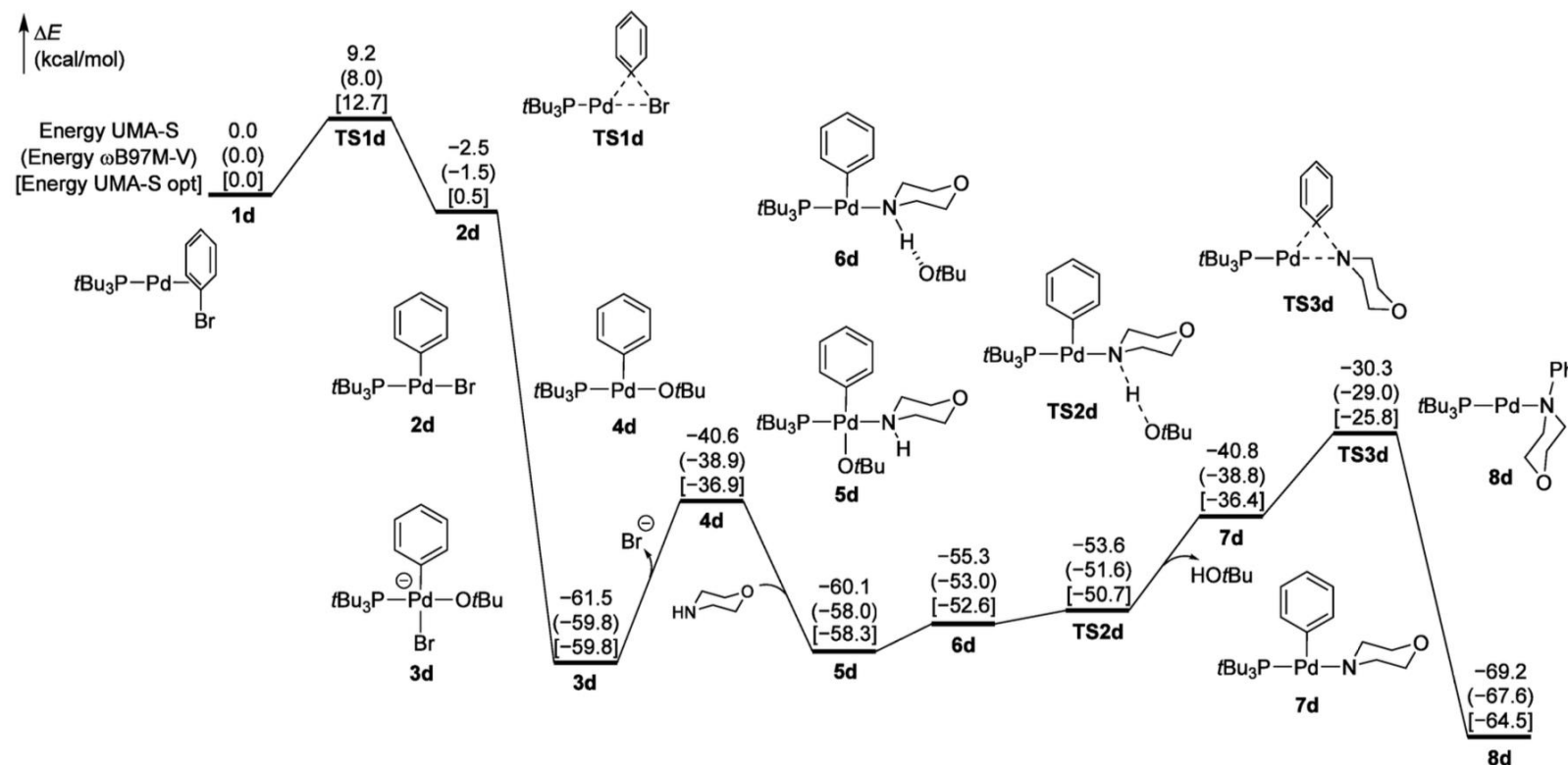
- ✓ Similar energies compared to ω B97M-V
- ✓ The mean absolute deviation:
 - UMA-S and ω B97M-V:
 - 1.7 kcal/mol (max. 2.3 kcal/mol)
 - UMA-M energies and ω B97M-V:
 - 0.6 kcal/mol (max. 1.0 kcal/mol)
- ✓ Activation barrier (UMA-S):
 - Oxidative-addition: <10 kcal/mol
 - reductive elimination: nearly 30 kcal/mol



UMA-S, UMA-M, and ω B97M-V single point energies, and UMA-S optimized energies

Cross-coupling Catalysis: Pd catalyzed Buchwald-Hartwig amination reaction

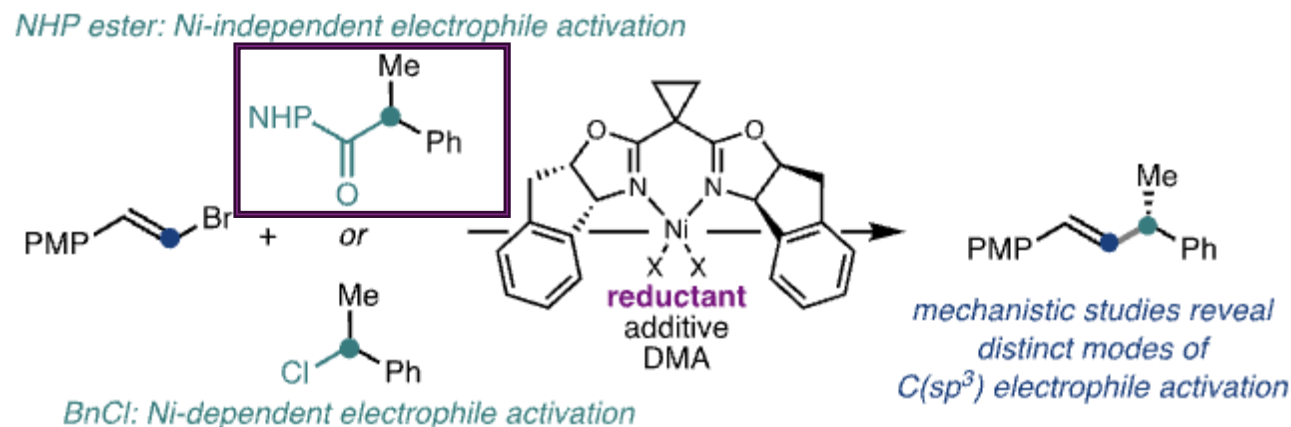
- Calculate anionic metal species (**3d**), outer-sphere reaction steps (**TS2d**).
- Coordination of the tert-butoxide strongly stabilizes the oxidized Pd metal center.
- Gas phase energies for coordination are close (differ by less than 2 kcal/mol).
- Reductive elimination can be the turnover-limiting bottleneck



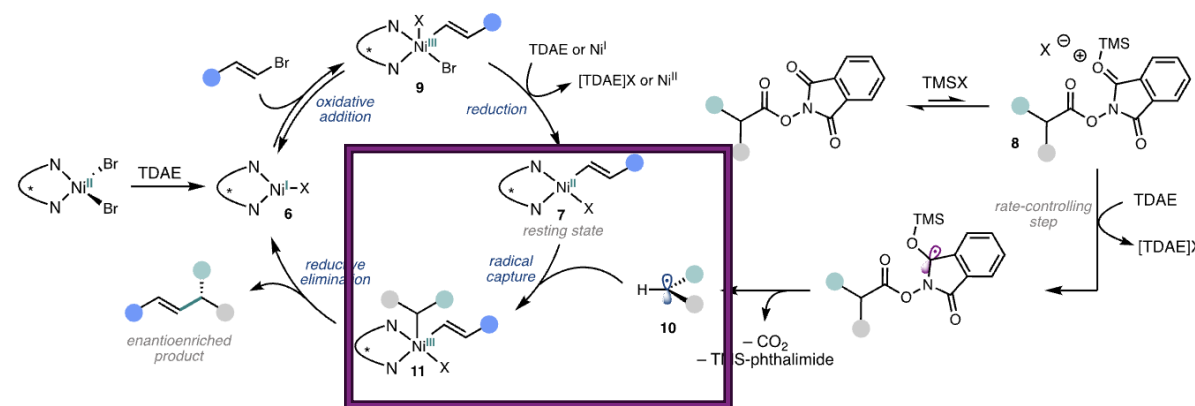
Energy profile for palladium-catalyzed Buchwald-Hartwig amination reaction

Case study 2: Asymmetric Catalysis

- ✓ Stereoselectivity of a Ni-catalyzed asymmetric reductive alkenylation of N-hydroxyphthalimide ester and benzylic chloride.
- ✓ Involvement of a Ni(II) to Ni(III) oxidation process through reaction of an open-shell carbon radical intermediate
- ✓ Original DFT evaluation:
 - M06/6-311+G(d,p)/SDD(Ni)/SMD(DMA)// B3LYP-D3/6-31G(d)/LANL2DZ(Ni)
 - Lowest energy route: S-product forming pathway
 - Gibbs energy difference for S-TS vs. R-TS: 3 kcal/mol
 - Enantiodetermining transition state: radical capture



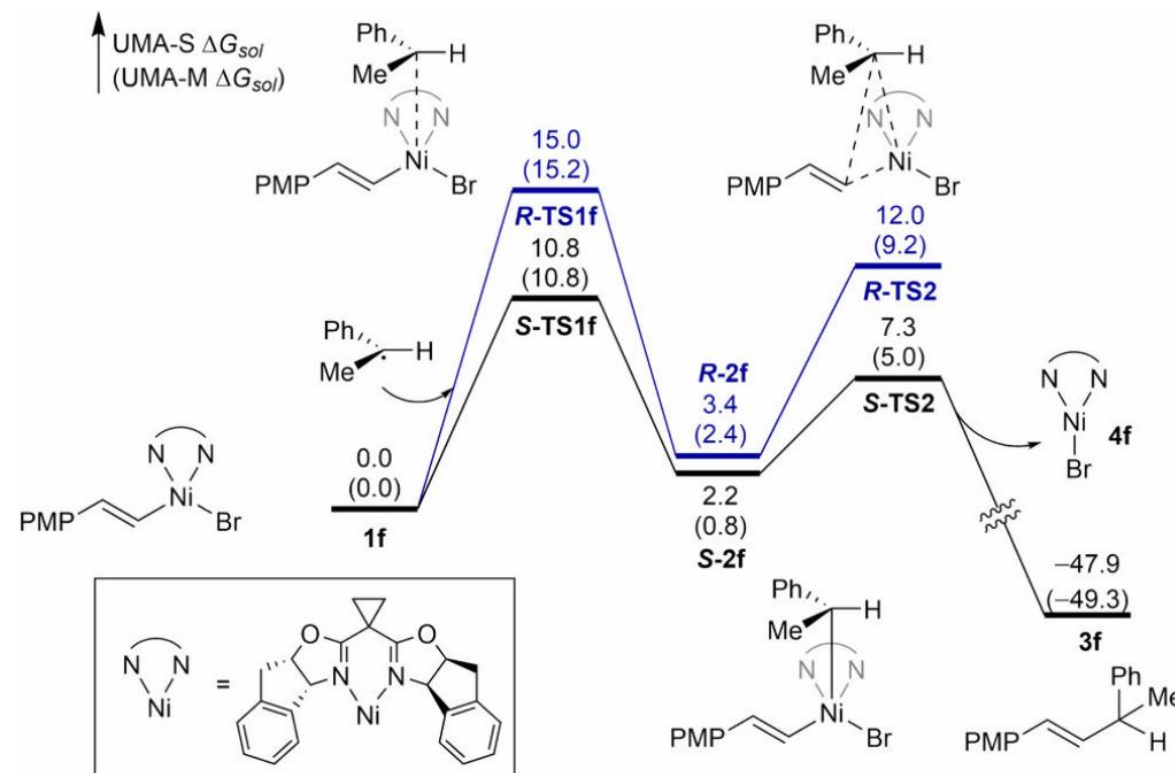
Overview of Ni-catalyzed asymmetric reductive alkenylation



Proposed mechanism for the TDAE-mediated Ni-catalyzed coupling

Asymmetric Catalysis

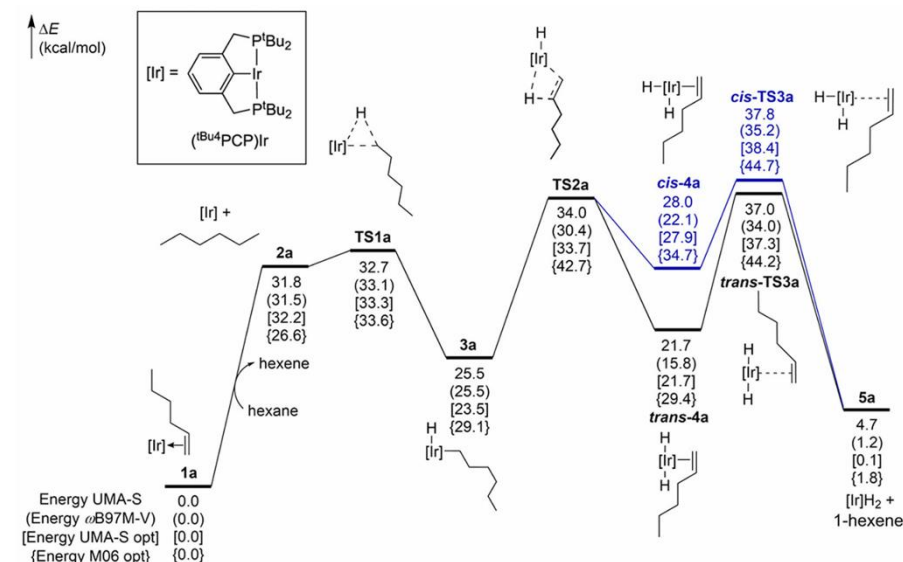
- ✓ UMA itself was not trained with an implicit-solvent model
- ALPB + GFN2-xTB
- UMA-S/ALPB(DMA) optimized structures
- UMA-M/ALPB(DMF)//UMA-S/ALPB(DMF) single point energies
- ✓ The global minimum structures were identified in less than 10 h.
- ✓ Lowest energy route: S-product pathway
- ✓ Stereodetermining transition state: radical capture
- Forward reductive elimination is easier
- ✓ Gibbs energy difference for S-TS1f vs. R-TS1f: 4.7 kcal/mol
- ✓ UMA-M provide similar qualitatively and quantitatively values



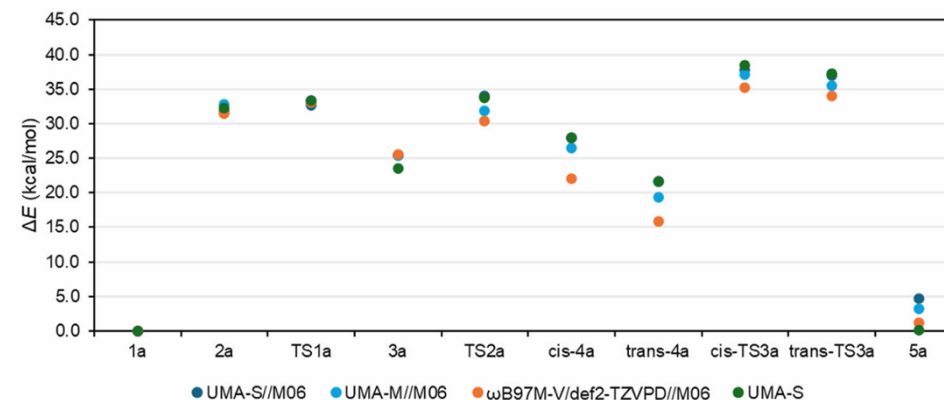
Energy profile for Ni-catalyzed enantioselective radical capture and reductive elimination

Case study 3: Alkane Dehydrogenation Catalysis

- ✓ Converts unreactive saturated hydrocarbons (alkanes) into valuable unsaturated building blocks (alkenes/olefins) and hydrogen gas:
- Unfavorable thermodynamics and slow kinetics.
- ✓ Introduction of (PCP)IrI pincer complex
- ✓ UMA-S gave the same energy span structures as M06
- ✓ UMA-S predicted 3a and 4a intermediates as relatively endothermic.
- ✓ UMA-S deviates from M06 by a few kcal/mol or each structure.
- ✓ UMA-M energies show a modestly better agreement



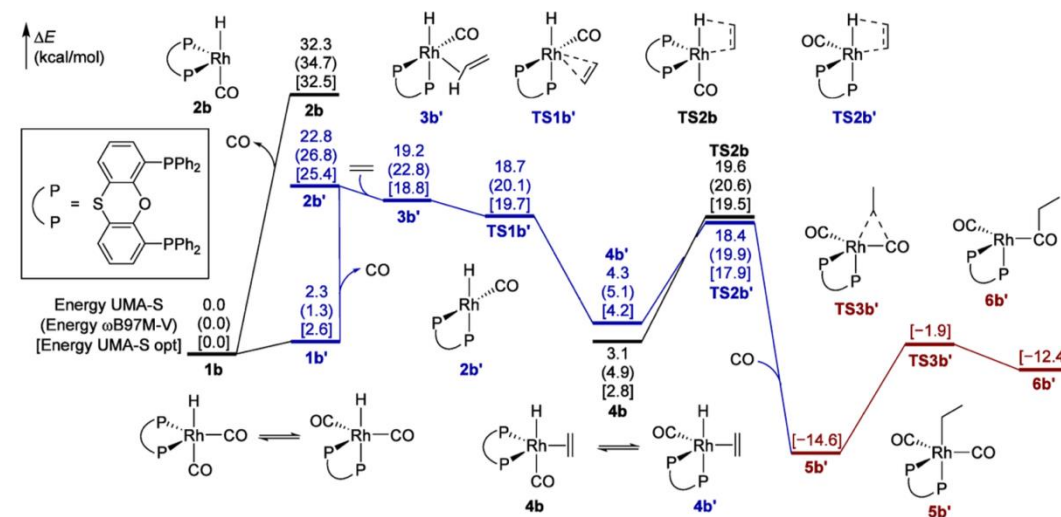
Energy profile for iridium-catalyzed hexane dehydrogenation



UMA-S, UMA-M, and ω B97M-V single point energies, and UMA-S optimized energies

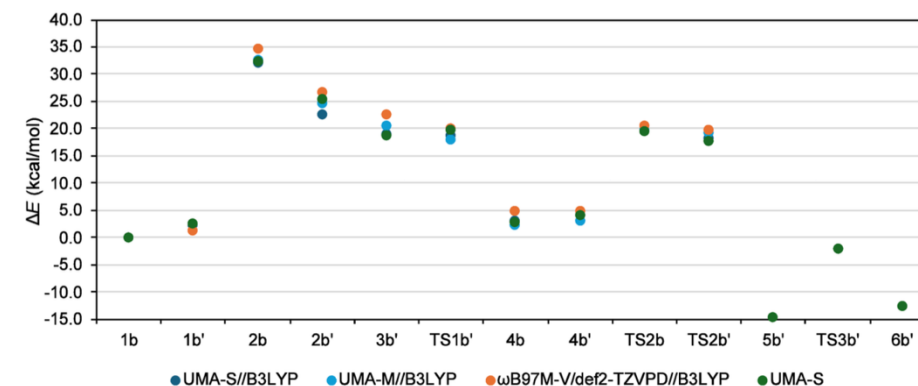
Case study 4: Hydroformylation Catalysis

- ✓ The conversion of alkenes to aldehydes using carbon monoxide and dihydrogen
- ✓ Good agreement between UMA-S, UMA-M, and ω B97M-V.
- Mean absolute error for single-point energies:
 - Between the UMA-S and ω B97M-V: 1.9 kcal/mol
 - Between the UMA-M and ω B97M-V : 1.6 kcal/mol
- ✓ UMA-S//UMA-S captured the relatively difficult initial carbon monoxide dissociation step:
- Energy of 25.4 kcal/mol
- Changes by <2 kcal/mol for ω B97M-V/def2-TZVPD.
- ✓ A higher energy intermediate can result in an overall lower energy pathway (1b, 2b vs. 1b', 2b')



Energy profile for rhodium-catalyzed hydroformylation

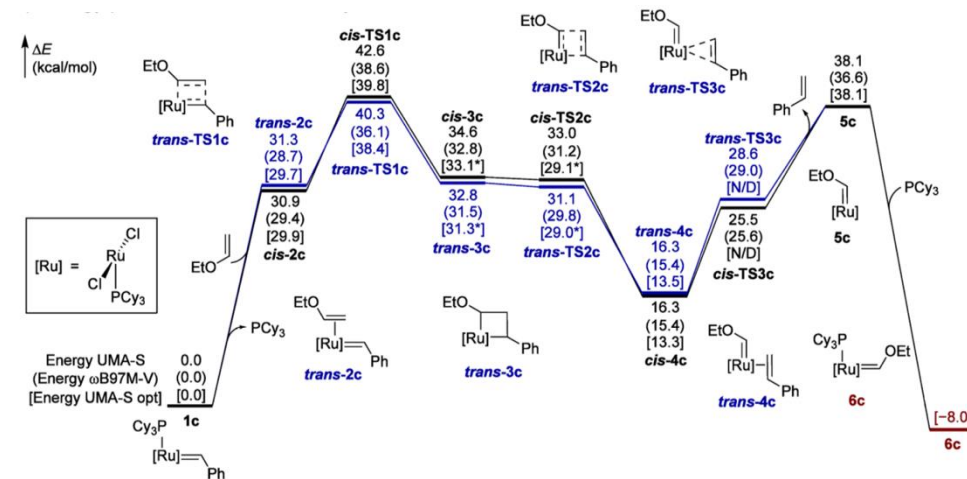
b) UMA-S, UMA-M, and ω B97M-V single point energies, and UMA-S optimized energies



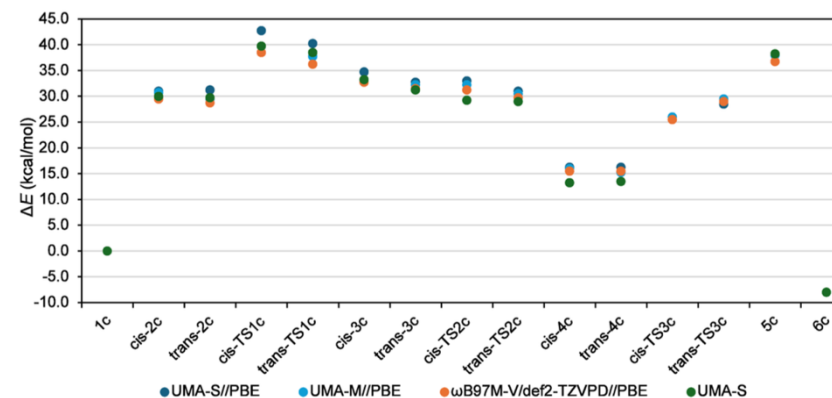
UMA-S, UMA-M, and ω B97M-V single point energies, and UMA-S optimized energies

Case study 5: Olefin Metathesis Catalysis

- ✓ Provides an opportunity to evaluate metal-carbene and four-membered metallacycle intermediates
- ✓ Small cis-trans energy difference between nearly all structures along the energy profile:
- The trans structure is lower in energy than the cis structure
- ✓ Mean absolute error for single-point energies:
 - Between the UMA-S and ω B97M-V: 1.7 kcal/mol
 - Between the UMA-M and ω B97M-V : 0.8 kcal/mol
- ✓ Optimization with UMA-S provided very similar results to UMA-S and UMA-M single point energies on geometries.

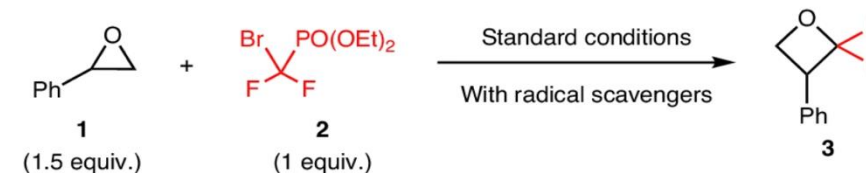


Energy profile for ruthenium-catalyzed olefin metathesis



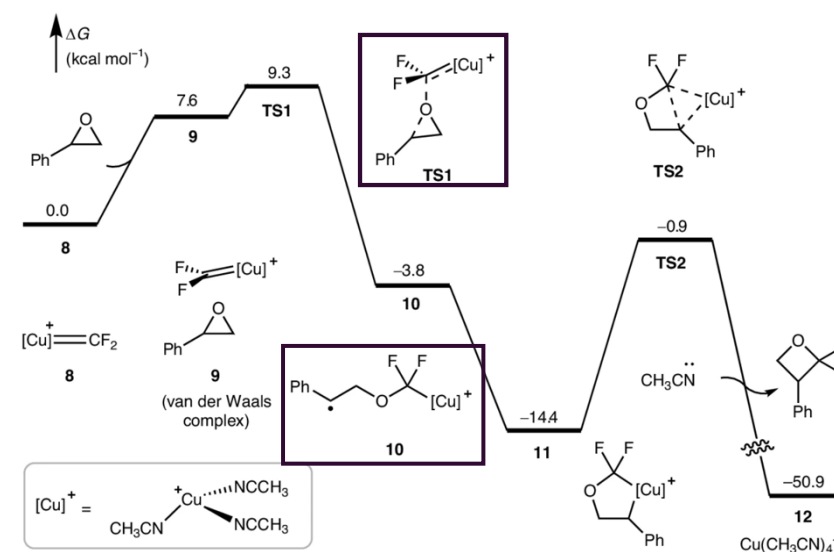
UMA-S, UMA-M, and ω B97M-V single point energies, and UMA-S optimized energies

Case study 6: Catalytic Cycle with an Open-shell Intermediate



Overview of difluorocarbene insertion

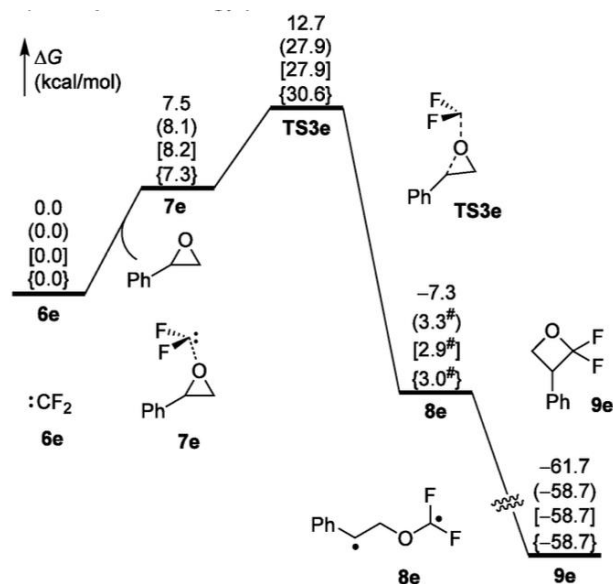
- ✓ Open-shell intermediate in OMol25 remain unclear.
- ✓ E.g. (Copper-catalyzed) difluorocarbene insertion into an epoxide.
 - Closed-shell transition state (**TS1**)
 - Open-shell singlet diradical intermediate (**10**)



Computed energy profile of the copper-catalyzed difluorocarbene insertion

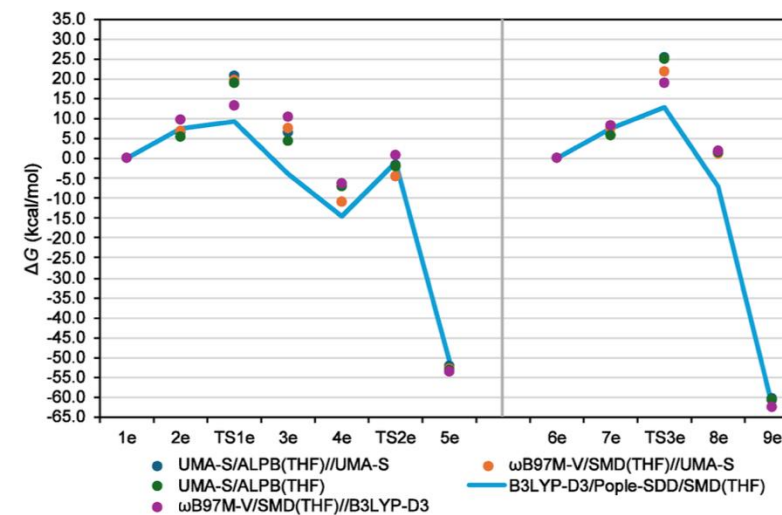
Copper-free difluorocarbene insertion into an epoxide and solvent effects

- ✓ 8e:
- Analogue to 3e with nearly degenerate singlet and triplet.
- ✓ Single point energies similar to ω B97M-V single:
- Absolute error: 0.4 kcal/mol
- Maximum error: 2.4 kcal/mol



Energy profile for difluorocarbene insertion

- ✓ Solvent effect:
- Correction: ALPB + GFN2-xTB
- ✓ Reasonable results with comparisons:
- Accuracy of UMA-S geometries
- ALPB as additive solvation corrections to UMA-S energies and geometry optimizations



Free energies of UMA-S and ω B97M-V with implicit solvation in THF

Copper-catalyzed difluorocarbene insertion into an epoxide

Unable to provide the optimized TS1e structure or an NEB transition state

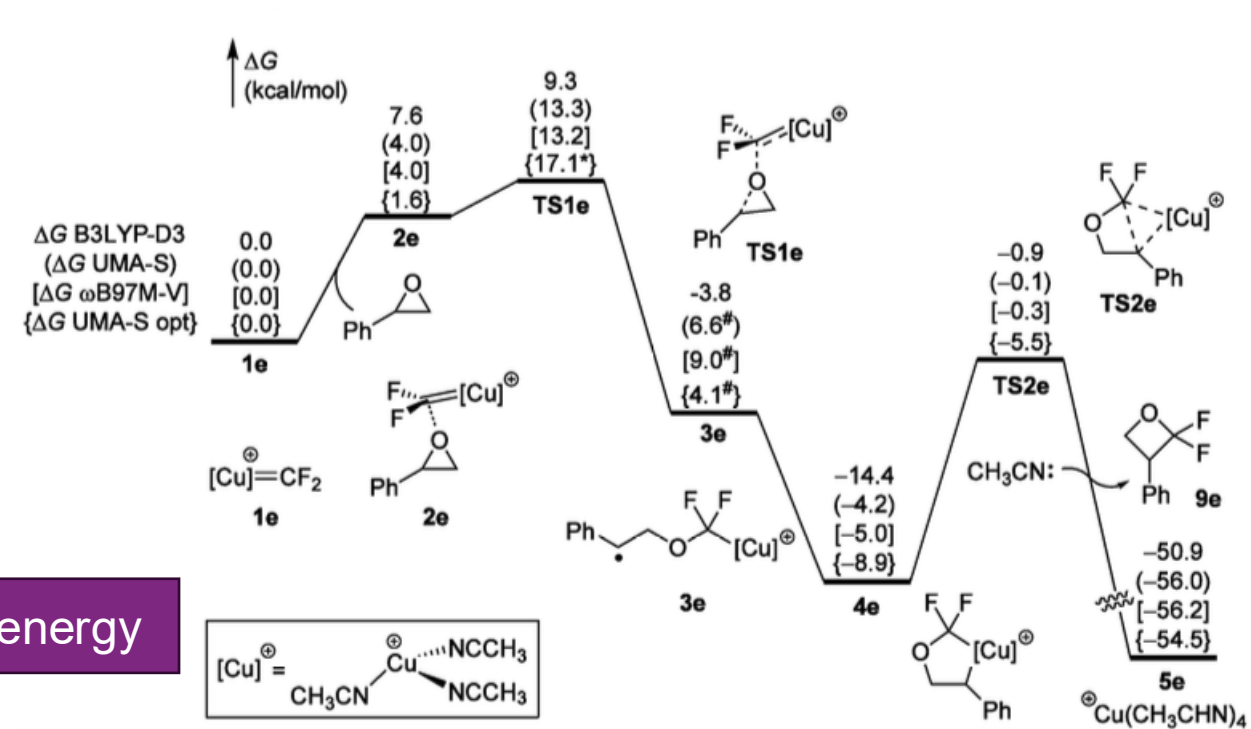
3e as a singlet: major unphysical conformational changes

3e: Lower triplet energy than singlet

Triplet energy: $\sim \omega$ B97M-V

not close to singlet energy

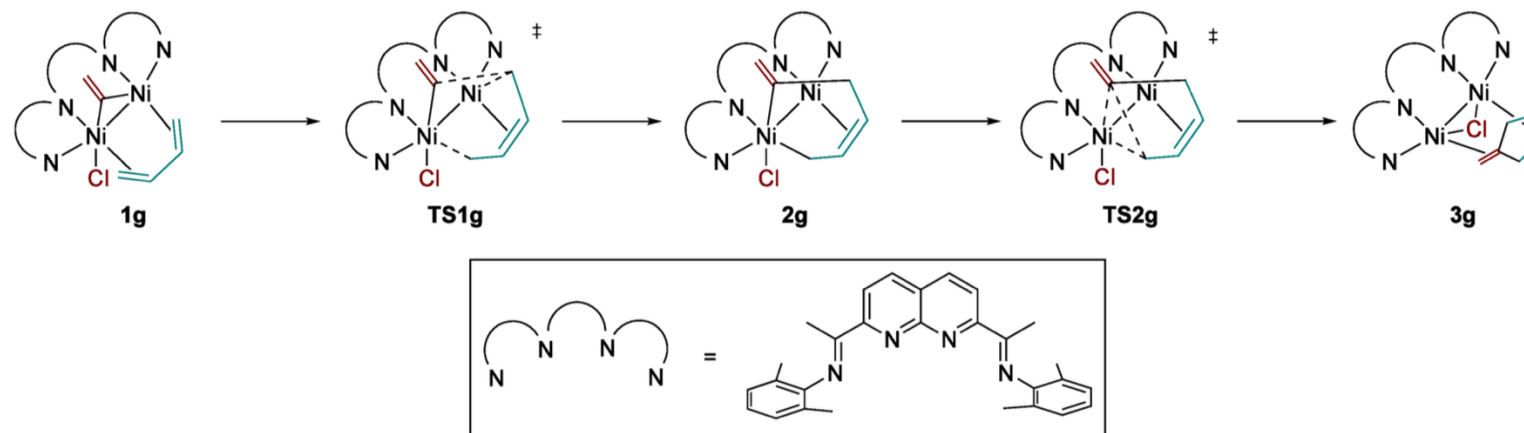
Weakness in structures and energies of organometallic open-shell singlet diradicals



Energy profile for palladium-catalyzed Buchwald-Hartwig amination reaction

Case study 7: Dinuclear Catalysis

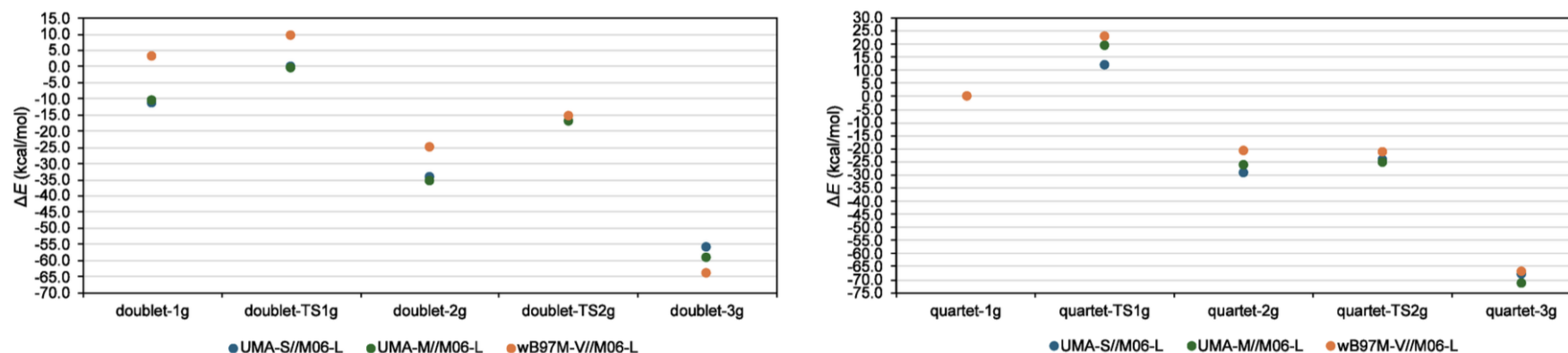
- ✓ Dinuclear Ni-Ni naphthyridine-diimine complex catalyzes the formal [4 + 1] cycloaddition of dienes



Proposed reaction pathway with 2D structures

Dinuclear Catalysis

- ✓ UMA reasonably captured the potential energy surface shapes for both the doublet and quartet spin states.



Comparison of electronic energies as the doublet (left) and quartet (right) spin states

- ✓ The OMol25 dataset did not include multinuclear transition metal complexes.
- ✓ UMA models struggled to produce reasonable ground state energies, especially for relative spin states:
- UMA did not predict that same spin state as ωB97M-V.
- ✓ UMA reproduced the qualitative energy profiles and optimized geometries, but its energy accuracy limits quantitative catalytic predictions.

Conclusion

- UMA successfully acts as a fast surrogate for ω B97M-V/def2-TZVPD DFT for a broad range of homogeneous transition-metal catalytic systems, including Ir, Rh, Ru, Pd, Cu and Ni chemistry.
- It can predict both geometries and relative energies of intermediates and transition states with generally good agreement with DFT.
- Cross the conventional closed-shell catalytic examples, UMA generally reproduces the overall shape of the DFT energy landscape, including chemically important features such as rate/selectivity-determining transition states.
- A major advantage is computational efficiency. UMA can perform geometry optimisation and energy evaluation far faster than conventional DFT, making large-scale tasks such as conformational searching, reaction-pathway exploration and catalyst screening much more practical.

Limitations and future development:

- Its performance deteriorates for challenging electronic structures, particularly open-shell singlets, spin-state energetics and systems with significant multireference character.
- All calculations using UMA-S or UMA-M will always be slightly less accurate than the ω B97M-V functional
- It is difficult to anticipate whether a given structure is within the chemical space captured by the UMA-S training data.
- it does not generate other properties such as atomic charges or spin densities.
- OMol25 only contains mononuclear structures UMA.

Thank You